

```
import seaborn as sns

df = sns.load_dataset('titanic')
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
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```
print("Rows and Columns:", df.shape)
```

Rows and Columns: (891, 15)

The dataset contains 891 rows and 15 columns.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male  891 non-null    bool
11  deck        203 non-null    category
12  embark_town 889 non-null    object
13  alive       891 non-null    object
14  alone       891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

Some columns such as age, deck, embarked, and embark_town contain missing values.

```
df.describe()
```

	survived	pclass	age	sibsp	parch	fare	
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000	
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208	
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429	
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000	
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400	
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200	
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000	
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200	

Average age is around 29 years. Maximum fare is approximately 512.

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

The deck column contains the highest number of missing values.

```
df['sex'].value_counts()
```

	count
sex	
male	577
female	314

dtype: int64

```
df['class'].value_counts()
```

	count
class	
Third	491
First	216
Second	184

dtype: int64

```
df['survived'].value_counts()
```

```
count
survived
0      549
1      342

dtype: int64
```

Most passengers were male and belonged to Third Class.

```
plt.figure(figsize=(8,5))

sns.histplot(df['age'], bins=20, kde=True)

plt.title("Age Distribution")
plt.show()
```

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_1719/4008984972.py in <cell line: 0>()
----> 1 plt.figure(figsize=(8,5))
      2
      3 sns.histplot(df['age'], bins=20, kde=True)
      4
      5 plt.title("Age Distribution")

NameError: name 'plt' is not defined
```

Next steps: [Explain error](#)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
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```

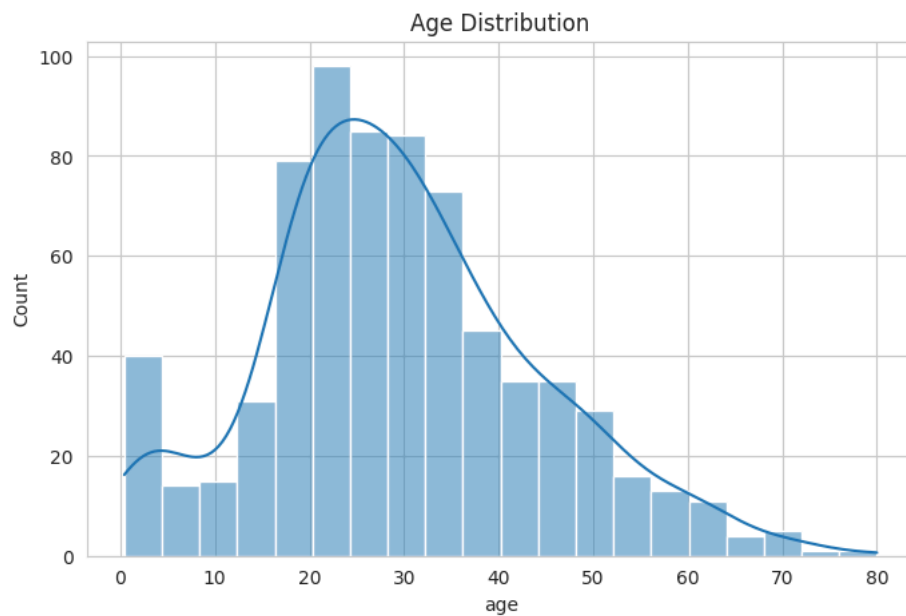
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
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```
plt.figure(figsize=(8,5))

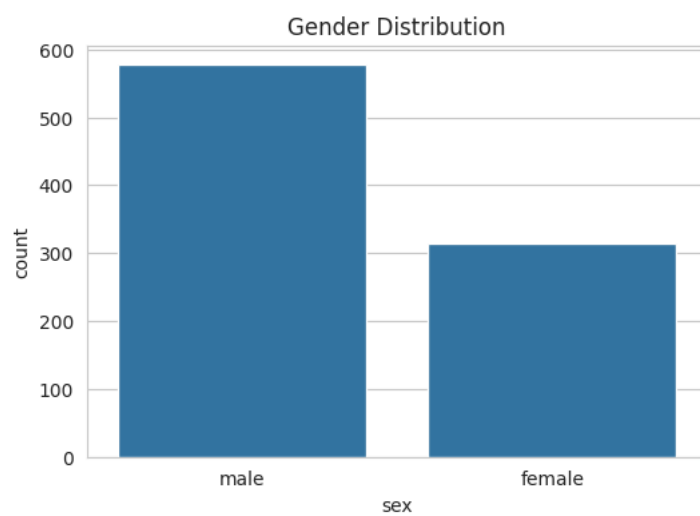
sns.histplot(df['age'], bins=20, kde=True)

plt.title("Age Distribution")
plt.show()
```



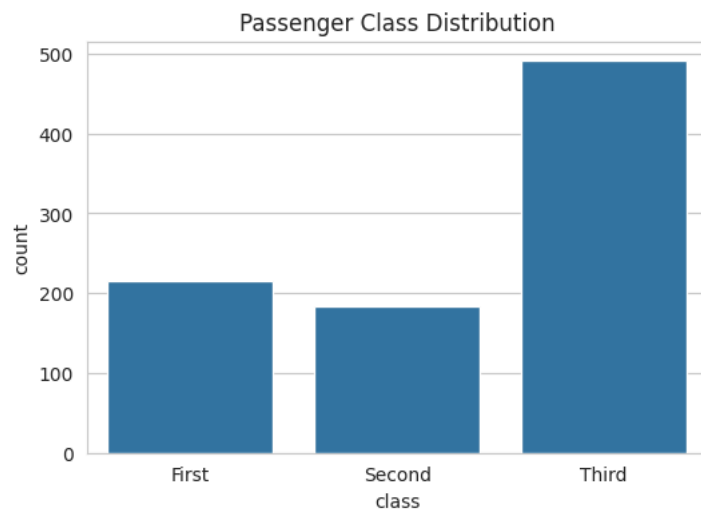
Most passengers were between 20 and 40 years old.

```
plt.figure(figsize=(6,4))  
  
sns.countplot(x='sex', data=df)  
  
plt.title("Gender Distribution")  
plt.show()
```



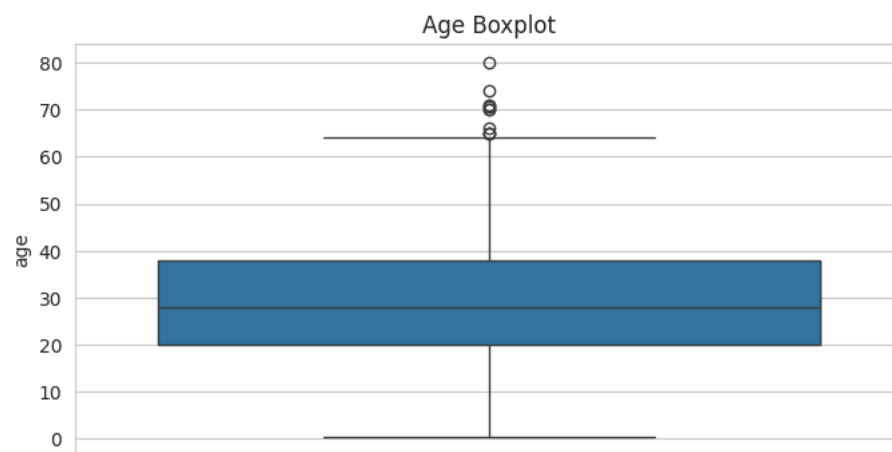
Male passengers were more numerous than females.

```
plt.figure(figsize=(6,4))  
  
sns.countplot(x='class', data=df)  
  
plt.title("Passenger Class Distribution")  
plt.show()
```



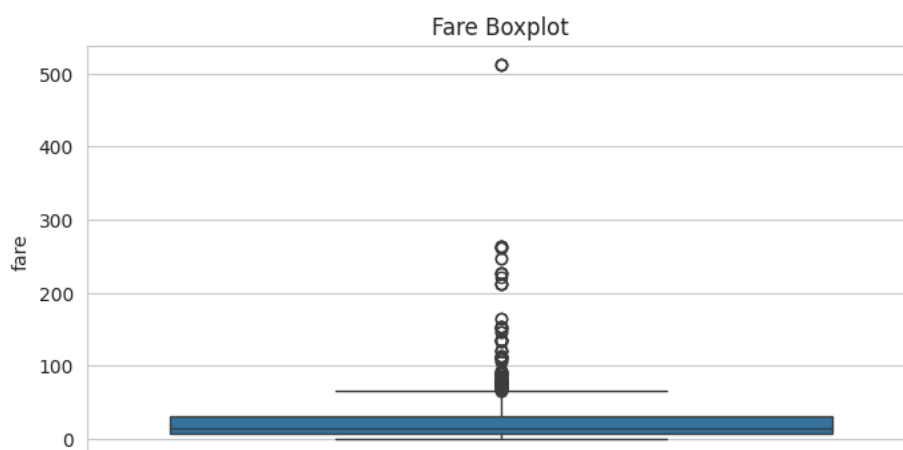
Third Class had the highest number of passengers.

```
plt.figure(figsize=(8,4))  
  
sns.boxplot(y=df['age'])  
  
plt.title("Age Boxplot")  
plt.show()
```



A few age outliers are present.

```
plt.figure(figsize=(8,4))  
  
sns.boxplot(y=df['fare'])  
  
plt.title("Fare Boxplot")  
plt.show()
```

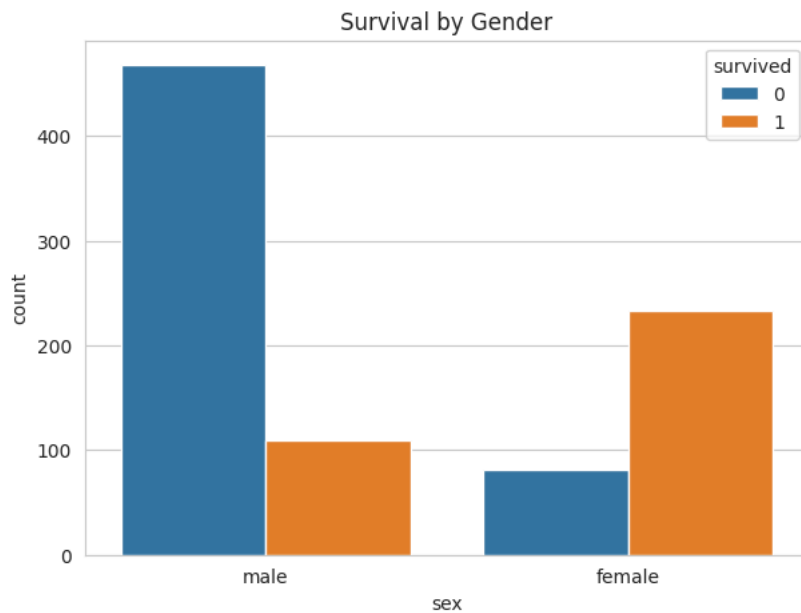


Several high-fare outliers are visible.

```
plt.figure(figsize=(7,5))

sns.countplot(
    x='sex',
    hue='survived',
    data=df
)

plt.title("Survival by Gender")
plt.show()
```

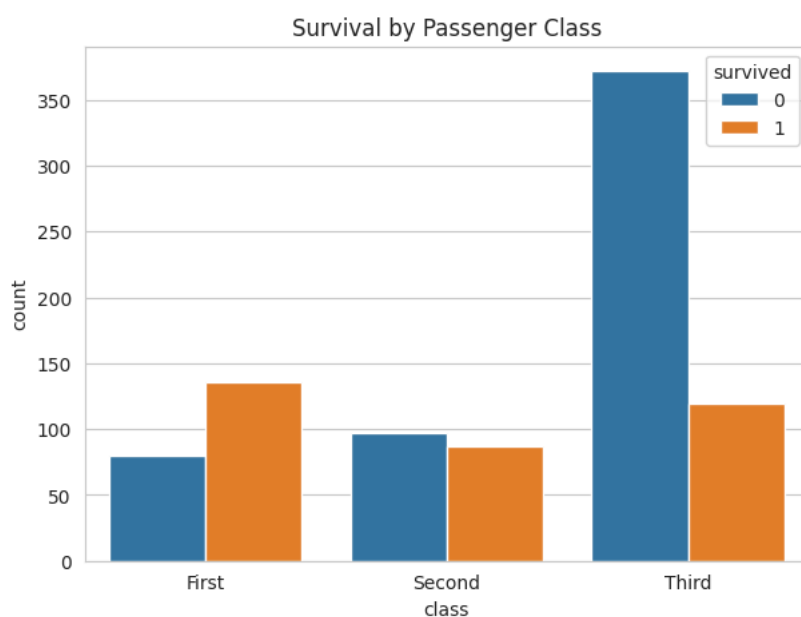


Female passengers had a much higher survival rate.

```
plt.figure(figsize=(7,5))

sns.countplot(
    x='class',
    hue='survived',
    data=df
)

plt.title("Survival by Passenger Class")
plt.show()
```

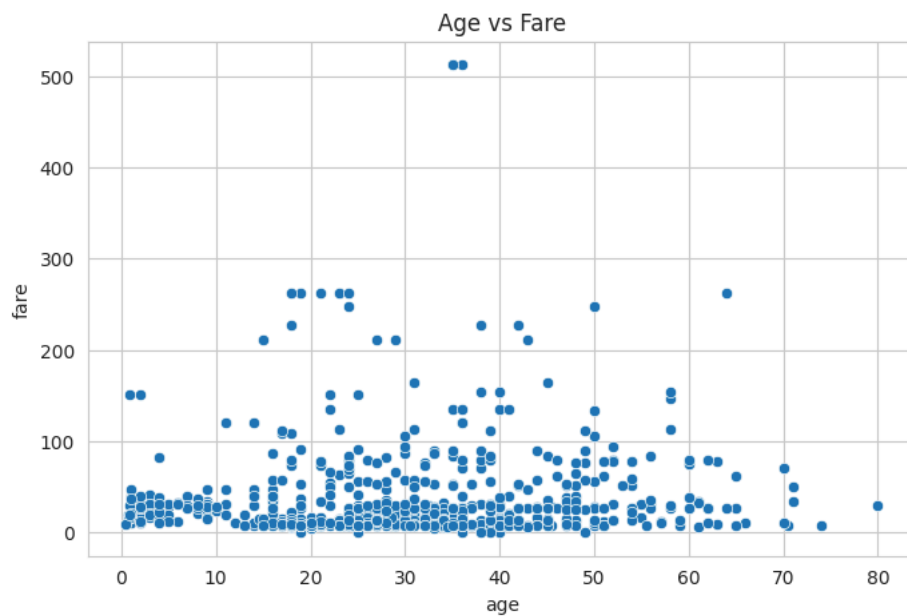


First-class passengers survived more frequently than Third-class passengers.

```
plt.figure(figsize=(8,5))

sns.scatterplot(
    x='age',
    y='fare',
    data=df
)

plt.title("Age vs Fare")
plt.show()
```



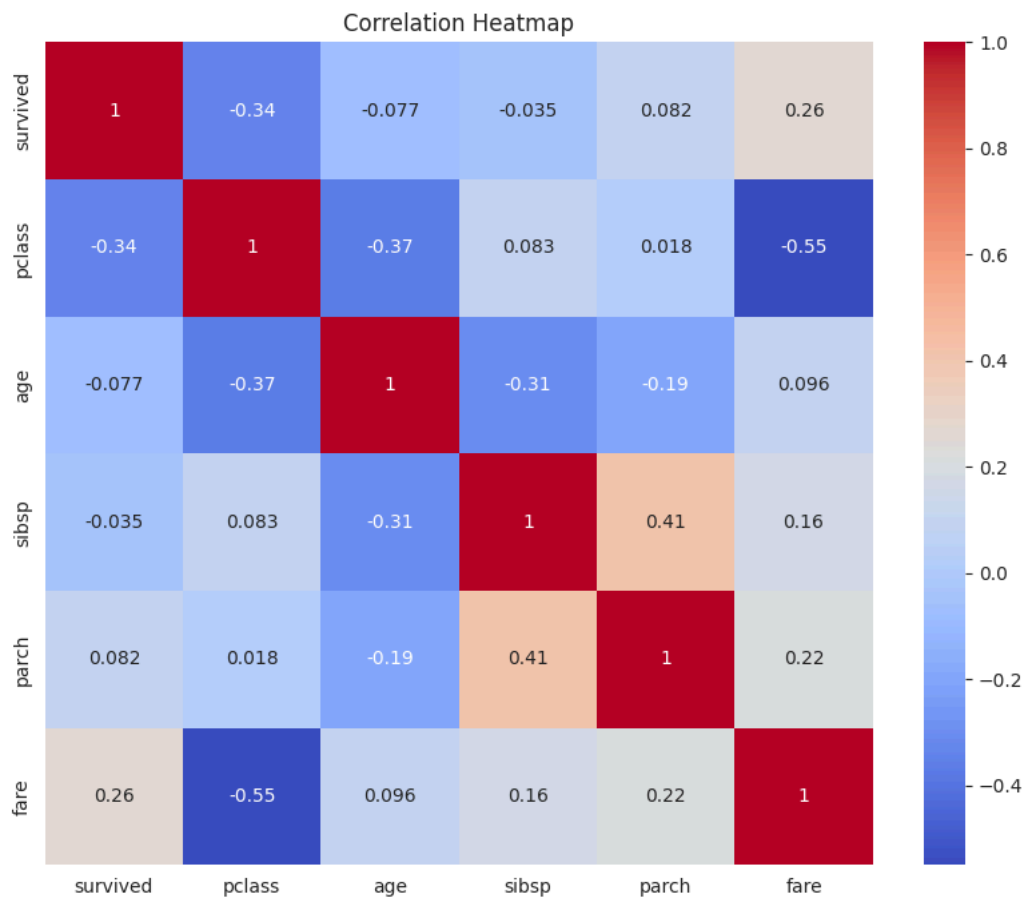
There is no strong relationship between age and fare.

```
numeric_df = df.select_dtypes(include=['number'])

plt.figure(figsize=(10,8))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")
plt.show()
```



Fare shows a positive correlation with survival. Passenger class is negatively correlated with survival.

```
sns.pairplot(  
    df[['age', 'fare', 'pclass', 'survived']]  
)  
  
plt.show()
```